



MATERIAL TEST REPORT
FFI-QA-CON-F0003
Revision: 08, 05/02/2017

Mill Test Report				
DNOW L.P.		Purchase Order:	66268198	
PO BOX 40985		Sales Order:	341638	
HOUSTON TX 77240		Line Item:	1	Qty: 4

Item Code:	02 24500 99	Material:	A105N LATEST EDITION	
Description:	4 900# RF BL FLANGE ANSI B16.5		Manufactured in the U.S.A.	
FID:	A7687	Heat Code:	M003010B	

Element		Ladle
C	Carbon	0.2000
Mn	Manganese	1.1900
P	Phosphorous	0.0110
S	Sulphur	0.0060
Si	Silicon	0.1900
Cu	Copper	0.0060
Ni	Nickel	0.0050
Cr	Chromium	0.0140

Element		Ladle
N	Nitrogen	
Mo	Molybdenum	0.0010
V	Vanadium	0.0040
CbNb	Columbium/Niobium	0.0030
Ti	Titanium	
Al	Aluminum	0.0260
B	Boron	0.0002
Sn	Tin	

Additional Chemicals	
NICRMOVCU 0.0300 CRMO 0.015	

Melting Practice	EAF LRF VD	Reduction Ratio	4:1
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Mechanical Test Results			
Tensile (PSI)	77,292		
Yield (PSI)	50,758		
Elong (%)	32.80		
Red (%)	72.91		
CE	0.4029		
PCM			
HBW	159	to	156
CVN Test Temperature	-50		°F
Ft/Lbs.	159	184	177
Shear (%)	70	70	70
MLE	72	75	71
Charpy Sample	10 X 10 X 55mm		
QTC Type & Size			

Heat Treatment Information	
Normalizing Temperature (°F)	1670
Normalizing Time (HRS)	5
Cooling Media (TYPE)	air

Additional Comments	
FULLY KILLED & FINE GRAIN PRACTICE	

	4/15/2025
ROLANDO PASTRAN - Quality Representative	Date

Product is compliant with NACE MR0175/ISO 15156 / Product is compliant with NACE MR0103. No weld repair is performed. The product is free of Mercury and Radiation contamination. Tensile test is taken from round specimen. Yield Strength is determined using 0.2% offset method, unless otherwise specified. When reported on the MTR: Tensile testing per ASTM E 8, Charpy impact testing per ASTM E 23 and Brinell Hardness Testing per ASTM E 10. Federal Flange manufactures products according to a certified QMS conforming to ISO 9001:2015. Products are manufactured in compliance to EN 10204 Type 3.1.	
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